Syna Malhan

Machine Learning • Data Science • Computer Vision • Cloud Analytics
Portfolio — synamalhan22@gmail.com — LinkedIn — GitHub

Education

Arizona State University *Expected May 2027*
B.S. Computer Science, Minor in Data Science GPA: 3.87/4.0
Honors: Dean's List (5x), Grace Hopper Celebration Scholar, NAMU Scholar

Technical Skills

Languages: Python, SQL, Java, C++, JavaScript
Machine Learning: CNNs, Transformers, GNNs, Clustering, Anomaly Detection, Time-Series Forecasting, Generative AI, RAG, Diffusion Models, Predictive Modeling
Data Science: EDA, Feature Engineering, Statistical Modeling, Predictive Analytics, Experimentation, Failure Analysis, Root Cause Analysis
Frameworks: PyTorch, Scikit-Learn, Pandas, NumPy, XGBoost, Neo4j
Cloud & Data: AWS (EC2, Lambda, DynamoDB), Snowflake, GCP, Azure, REST APIs, SAP

Experience

Micron Technology — Data Science Intern *May 2026 – Aug 2026*

- Developed machine learning and AI solutions supporting semiconductor product validation, manufacturing optimization, and engineering decision-making.
- Built scalable analytics workflows in Snowflake and cloud environments (GCP, Azure) to process and analyze large-scale failure analysis datasets.
- Applied clustering and unsupervised learning techniques to identify defect signatures, anomaly patterns, and root-cause relationships across manufacturing processes.
- Engineered features and statistical analysis pipelines to improve process monitoring, anomaly detection, and product quality insights.
- Collaborated with cross-functional engineering teams to translate domain challenges into deployable data science solutions.

MPS Lab, Arizona State University — Undergraduate Researcher *Aug 2025 – Present*

- Conduct research in Intelligent Transportation Systems (ITS), spatio-temporal machine learning, and neural network architectures for traffic incident detection.
- Extended IncidentNet by integrating physics-informed constraints, improving model generalization under low-observability conditions.
- Developed synthetic and real-world data pipelines simulating heterogeneous sensor coverage, sparse observations, and missing-data scenarios.
- Evaluated model robustness across unseen transportation network topologies and varying sensor distributions.
- Contributed to experimentation, model development, and research validation for publication-oriented projects.

Ernst & Young (EY) — ML & Generative AI Intern *Jun 2025 – Aug 2025*

- Built enterprise-grade Retrieval-Augmented Generation (RAG) systems for structured and unstructured financial credit datasets.
- Developed vector-search pipelines and graph-based knowledge retrieval systems using Neo4j and embedding models.
- Improved retrieval relevance and reasoning performance through prompt engineering, embedding optimization, and evaluation workflows.
- Designed AI solutions capable of extracting insights across large financial document repositories.
- Developed text-to-image generation models with improved semantic alignment and generation quality.

Infinite Uptime — Data Science Intern *Jul 2025 – Sep 2025*

- Analyzed millions of industrial IoT sensor readings from rotating equipment and bearing systems.
- Engineered temporal, frequency-domain, and multivariate features to capture degradation trends and failure indicators.
- Developed predictive machine learning models forecasting equipment failures days in advance to support predictive maintenance strategies.
- Delivered interpretable model outputs and failure explanations enabling data-driven maintenance planning.

ASU AI Cloud Innovation Center — Cloud Developer *Aug 2024 – Aug 2025*

- Designed and deployed multiple production-grade AI applications serving academic and industry stakeholders.
- Built full-stack applications integrating frontend systems with AWS-hosted APIs, DynamoDB databases, and cloud services.
- Developed scalable cloud infrastructure using EC2, Lambda, and serverless architectures supporting real-time inference workloads.
- Standardized deployment workflows and reusable cloud templates to accelerate development cycles.

Ripik AI — ML & Computer Vision Intern *Jul 2024 – Sep 2024*

- Improved industrial slab-sizing accuracy from 76% to 97% using computer vision and image-processing pipelines.
- Corrected camera distortion and improved model robustness through geometric calibration and targeted augmentation techniques.
- Developed OCR systems for low-quality warehouse imagery, automating material identification and inventory workflows.

Jindal Steel Ltd. (JSPL) — Digitalization & ML Intern *Jun 2024 – Aug 2024*

- Developed machine learning models predicting finished-goods production using large-scale manufacturing datasets.
- Improved production planning efficiency by approximately 80% through predictive analytics and process optimization.
- Designed logistics optimization systems reducing unused rake capacity and lowering transportation costs by roughly 60%.
- Built internal digital platforms integrated with SAP systems and deployed on enterprise infrastructure.

Headstarter AI — Software Engineering Fellow *Aug 2024 – Oct 2024*

- Built production-focused software solutions in collaborative hackathon environments.
- Worked with engineers from Amazon and Google on software architecture, development workflows, and code reviews.
- Ranked Top 5 among 500+ participating teams based on technical execution and project quality.

Projects

Wildlife Monitoring & Endangered Species Protection

- Developed computer vision models to detect and classify wildlife species from large-scale camera trap datasets.
- Improved robustness to motion blur, occlusion, and low-light environmental conditions.
- Designed scalable monitoring pipelines supporting wildlife conservation, species tracking, and population analysis.

Graph-Based Recommendation System

- Modeled user-item interactions as graph structures and trained graph-based recommendation models.
- Leveraged graph learning techniques to improve recommendation relevance compared to collaborative filtering baselines.
- Evaluated recommendation performance using ranking and relevance metrics.